MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

Topic 4: Angular Momentum and Central Forces - Problem Sets

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Source: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

OFFICIAL MATERIAL Problems.pdf · p. 9

Statements and solutions: Each card gives the statement from the official collection (with the PDF page) and a worked solution in four phases: hypotheses and data, frames and change-of-basis matrices, step-by-step derivation, and interpretation with unit and limit checks. Only Problems 17 and 39 (part f) have an official key entry; the other results are derived here and checked numerically.

PROBLEM 17 Central Force + Gravity

Linear Central Force under Gravity

Original statement · Problems.pdf, p. 9

A force attracts a heavy particle M of mass m to a point O. The attractive force is proportional to the mass of the particle and to the distance OM. The proportionality coefficient is k^2. The coordinate system $Oxyz$ has the origin at the centre of attraction O and the axis Oz is vertically oriented. Initially, the coordinates of M are

$$(x_0, y_0, z_0) = \left(\frac{3g}{k^2}, 0, 0 \right)$$

and its initial velocity is

$$(\dot{x}_0, \dot{y}_0, \dot{z}_0) = \left(0, \frac{2g}{k}, 0 \right)$$

(a) Describe the motion of M, specifying clearly the shape of the path.

(b) What is the velocity of the particle as a function of t?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a heavy particle M of mass m under (i) a linear central attraction $\mathbf{F}_a = -mk^2\mathbf{r}$ towards O and (ii) its weight $\mathbf{F}_g = -mg\mathbf{k}$ (Oz points up).
- **Hypotheses:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is inertial; no drag or friction; m, g, k are positive constants.
- **Degrees of freedom:** free particle in 3D space, $\mathrm{CDOF} = 3$ with coordinates (x, y, z).
- **Initial conditions ($t = 0$):** $\mathbf{r}(0) = \dfrac{3g}{k^2}\mathbf{i}$ and $\mathbf{v}(0) = \dfrac{2g}{k}\mathbf{j}$.

Symbol	Meaning	SI unit
m	mass of M	kg
k	attraction coefficient is k^2	s^{-1}
g	gravitational acceleration	$\mathrm{m/s}^2$
g/k^2	length scale of the problem	m
g/k	speed scale of the problem	m/s

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathcal{B}_0\}$, $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$; $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}$.
- **Orbit-plane basis** (introduced after the planar result of Phase 3): $\mathcal{B}_1 = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ with

$$\mathbf{e}_1 = \left(\tfrac{3}{2}, 0, \tfrac{1}{2}\right), \qquad \mathbf{e}_2 = \mathbf{j}, \qquad \mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2 = \left(-\tfrac{1}{2}, 0, \tfrac{3}{2}\right)$$

Change-of-basis matrix (columns are the components of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} 3/2 & 0 & -1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 3/2 \end{pmatrix}$$

Checks. $\mathbf{e}_1 \cdot \mathbf{e}_1 = \tfrac{3}{4} + \tfrac{1}{4} = 1$, $\mathbf{e}_3 \cdot \mathbf{e}_3 = \tfrac{1}{4} + \tfrac{3}{4} = 1$, $\mathbf{e}_1 \cdot \mathbf{e}_3 = -\tfrac{3}{4} + \tfrac{3}{4} = 0$, hence $[{}_0R_1]^T[{}_0R_1] = I$; and $\det[{}_0R_1] = \tfrac{3}{4} + \tfrac{1}{4} = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

1. Equations of motion

Pedagogical Justification: $m\mathbf{a} = \mathbf{F}_a + \mathbf{F}_g$ in the inertial frame. Because the restoring force is linear in $\mathbf{r}$, each Cartesian component decouples.

$$m\left(\ddot{x}\mathbf{i}+\ddot{y}\mathbf{j}+\ddot{z}\mathbf{k}\right)=-mk^2\left(x\mathbf{i}+y\mathbf{j}+z\mathbf{k}\right)-mg\mathbf{k}\implies \ddot{x}+k^2x=0,\quad \ddot{y}+k^2y=0,\quad \ddot{z}+k^2z=-g$$

2. Integration of $x(t)$ and $y(t)$

> **Pedagogical Justification:** $\ddot{q}+k^2q=0$ is linear with constant coefficients; its characteristic roots are $\lambda=\pm ik$.

$$x(t)=A_1\cos kt+B_1\sin kt,\qquad \dot{x}(t)=-kA_1\sin kt+kB_1\cos kt$$

Initial data $x(0)=\quad 3g/k^2, \dot{x}(0)=0$ give $A_1=\quad 3g/k^2$ and $kB_1=0$. Likewise $y(0)=0\Rightarrow A_2=0$ and $\dot{y}(0)=2g/k\Rightarrow kB_2=2g/k\Rightarrow B_2=2g/k^2$:

$$x(t)=\frac{3g}{k^2}\cos kt,\qquad y(t)=\frac{2g}{k^2}\sin kt$$

3. Integration of $z(t)$

> **Pedagogical Justification:** The equation $\ddot{z}+k^2z=-g$ is forced by a constant. A constant particular solution z_p satisfies $k^2z_p=-g$, i.e. $z_p=-g/k^2$ (the equilibrium height), and the homogeneous part is as before.

$$z(t)=-\frac{g}{k^2}+A_3\cos kt+B_3\sin kt,\qquad \dot{z}(t)=-kA_3\sin kt+kB_3\cos kt$$

$z(0)=0\Rightarrow A_3=g/k^2$ and $\dot{z}(0)=0\Rightarrow B_3=0$:

$$z(t)=\frac{g}{k^2}\left(\cos kt-1\right)$$

4. Shape of the path

Planar confinement. x and z share the phase function $\cos kt$: from $\cos kt=k^2x/(\quad 3g)$,

$$z = \frac{g}{k^2}\left(\frac{k^2 x}{3g} - 1\right) = \frac{x}{3} - \frac{g}{k^2} \implies x - \quad 3\,z - \frac{3g}{k^2} = 0$$

The plane has normal $\mathbf{n} = (1, 0, -\quad 3)/2$ and contains the direction $(\quad 3, 0, 1)$, which makes $\arctan(1/\quad 3) = 30^\circ$ with the horizontal: the plane is inclined 30° to the horizontal.

Circle. With $\mathbf{C} = (0, 0, -g/k^2)$, the distance to $\mathbf{C}$ is constant:

$$x^2 + y^2 + \left(z + \frac{g}{k^2}\right)^2 = \frac{g^2}{k^4}\left[3\cos^2 kt + 4\sin^2 kt + \cos^2 kt\right] = \frac{4g^2}{k^4}$$

The distance from $\mathbf{C}$ to the plane is $\left|0 - \quad 3(-g/k^2) - \quad 3g/k^2\right| / \quad 4 = 0$, so the plane passes through the centre of the sphere and the path is a **great circle of radius** $R_c = 2g/k^2$, covered at angular rate k. In the plane basis of Phase 2 the three solutions combine into one vector identity:

$$\mathbf{r}(t) - \mathbf{C} = R_c\left[\cos kt\,\mathbf{e}_1 + \sin kt\,\mathbf{e}_2\right]$$

since $R_c \cos kt\,\mathbf{e}_1 = \left(\frac{3g}{k^2}\cos kt,\ 0,\ \frac{g}{k^2}\cos kt\right)$ reproduces $x(t)$ and $z(t) + g/k^2$, and $R_c \sin kt\,\mathbf{e}_2$ reproduces $y(t)$. The $\mathbf{e}_3$-component vanishes (planar motion) and $\mathbf{e}_3 \cdot \mathbf{k} = \quad 3/2$ confirms the 30° inclination of the normal to the vertical. Only the projection on Oxy, $\left(\frac{x}{3g/k^2}\right)^2 + \left(\frac{y}{2g/k^2}\right)^2 = 1$, is an ellipse; the circle is seen obliquely.

5. Velocity (part b)

> **Pedagogical Justification:** Differentiate each component with respect to time (chain rule $\frac{d}{dt}\cos kt = -k\sin kt$).

$$\mathbf{v}(t) = -\frac{3g}{k}\sin kt\,\mathbf{i} + \frac{2g}{k}\cos kt\,\mathbf{j} - \frac{g}{k}\sin kt\,\mathbf{k}$$

$$v^2 = \frac{g^2}{k^2}\left[3\sin^2 kt + 4\cos^2 kt + \sin^2 kt\right] = \frac{4g^2}{k^2}\left(\sin^2 kt + \cos^2 kt\right) = \frac{4g^2}{k^2} \implies v = \frac{2g}{k} = \text{const}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Why the speed is constant.** The force is conservative with potential $V = \frac{1}{2}mk^2(x^2 + y^2 + z^2) + mgz = \frac{1}{2}mk^2\left[x^2 + y^2 + (z + g/k^2)^2\right] - \dfrac{mg^2}{2k^2}$, which depends only on the distance to $\mathbf{C}$. That distance is constant on the circle, so V and therefore the kinetic energy are constant; also $v = kR_c = 2g/k$.
- **Initial-condition check:** $\mathbf{r}(0) = (\quad 3g/k^2, 0, 0)$ and $\mathbf{v}(0) = (0, 2g/k, 0)$ are reproduced by the formulas above.
- **Limit $g \to 0$:** the centre $\mathbf{C} \to O$ and the circle lies in a plane through O, but the radius $R_c = 2g/k^2 \to 0$ as well (the initial data $\quad 3g/k^2$ and $2g/k$ also vanish), so the path shrinks to the point O; it is not an elliptic oscillator orbit of finite size.
- **Dimensions:** $[g/k^2] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}}} = \mathrm{m}$, $[g/k] = \mathrm{m/s}$, $[kt] = 1$.
- **Numerical check (Python, $g = 9.81$, $k = 2$):** over 20 s, $\max\left|\|\mathbf{r} - \mathbf{C}\| - R_c\right| = 9 \times 10^{-16}$ m, $\max|v - 2g/k| = 2 \times 10^{-15}$ m/s, the plane residual $x - \quad 3z - \quad 3g/k^2$ stays below 3×10^{-15} m, and the finite-difference residual of $\ddot{x} + k^2 x = 0$ and $\ddot{z} + k^2 z + g = 0$ is below 4×10^{-7}.
- **Official key** (Problems.pdf, PDF page 73): circle given by the sphere $x^2 + y^2 + (z + g/k^2)^2 = 4g^2/k^4$ and the plane $x - \quad 3z = \quad 3g/k^2$, constant speed $2g/k$.

Worked Solution

Source: Problems.pdf (PDF pages 9-10); key on PDF page 73

PROBLEM 39

String + Edge Constraint

Particle on a String over a Plate Edge

Original statement · Problems.pdf, p. 27

A heavy point particle P with mass m is connected with point O on the upper surface of a thin square plate by a massless, inextensible string of length ℓ. The distance between O and the edge of the plate is d. The string hangs over the edge of the plate and can slide without friction. You can assume that the string remains tense and in contact with the plate at all times.

We denote the geometric point where the string is in contact with the plate edge as Q. We define the inertial reference frame $S_0 : \{O, \mathcal{B}_0\}$, with $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ as shown in the figure (Problems.pdf, PDF page 27). We define the angle θ between the Ox_0 axis and the string segment OQ, and the angle ϕ between a vertical plane containing the plate edge and the line perpendicular to the plate edge that passes through P. Initially at time $t = 0, \theta(0) = \phi(0) = \pi/3$ rad, and the modulus of the velocity of P is $v_0^P(0) = \quad \ell g$, where g is the magnitude of the acceleration due to gravity.

Hint: Note that under the conditions of the problem, the two angles marked with dashed lines are equal.

(a) How many degrees of freedom does the system have?

(b) Obtain the general expression of a unit vector in the direction from Q to P as a function of θ and ϕ.

(c) Obtain the general expression for the position, velocity, and acceleration vectors of Q and P in S_0 as a function of θ, ϕ, their derivatives, and the parameters of the problem.

(d) Identify and express mathematically all the forces acting on P.

(e) Find a set of second order differential equations to determine the evolution of θ and ϕ in time, free of any other unknowns.

(f) Does the tension force exert work on P? Provide proof of your answer.

(g) Derive the general expression of the mechanical energy of P in S_0 and compute its value initially. Is this a conserved quantity of motion? Justify your answer.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** heavy particle P on a massless string of length ℓ that slides without friction over the straight edge of a plate; Q is the contact point.
- **Hypotheses:** S_0 is inertial and $\mathbf{g} = -g\mathbf{k}_0$ (Oz_0 up). The edge is the line $x_0 = d$, $z_0 = 0$, parallel to $\mathbf{j}_0$ (read from the figure). The edge is smooth and the string is massless, so the tension $T \geq 0$ is the same on both sides of Q and the edge force on the string is normal to the edge. P does not touch the plate. No drag.
- **Degrees of freedom:** P has 3 coordinates and one holonomic constraint $\|\mathbf{OQ}\| + \|\mathbf{QP}\| = \ell$, so $\mathrm{CDOF} = 3 - 1 = 2$ with generalized coordinates (θ, ϕ).
- **Initial state:** $\theta(0) = \phi(0) = \pi/3$ and $\|\mathbf{v}_0^P(0)\| = \quad \ell g$. The statement gives only the speed, not its direction; for numerical checks $\dot{\phi}(0) = 0$ is assumed (the initial energy does not depend on it).

Symbol	Meaning	SI unit	Value in the numerical check
m	mass of P	kg	1
ℓ	string length	m	2.0
d	distance from O to the edge	m	0.4
g	gravitational acceleration	m/s^2	9.81

At $t = 0$ the hanging length is $\ell - d/\cos(\pi/3) = \ell - 2d$, so the data must satisfy $\ell > 2d$.

Assumptions and absence of an official key. The official key (Problems.pdf, PDF page 76) answers only part (f). Parts (a)-(e) and (g) are derived here from the statement and the figure on PDF page 27, with two explicit assumptions: the edge is the line $x_0 = d$ parallel to $\mathbf{j}_0$, and ϕ is measured as described in the statement. For those parts

there is no official result to compare with, so no discrepancy can be reported; the result for (f) agrees with the key.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Contact point Q.** It lies on the edge with OQ at angle θ from Ox_0:

$$\mathbf{r}_0^Q = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0, \qquad \|\mathbf{OQ}\| = \frac{d}{\cos\theta}, \qquad \hat{\mathbf{u}}_{OQ} = \cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0$$

Hanging length: $\rho \equiv \|\mathbf{QP}\| = \ell - d/\cos\theta$. Define $w \equiv \rho\cos\theta = \ell\cos\theta - d > 0$.

Why the dashed angles are equal. The string element at Q is massless, so the forces on it add to zero: $-T\hat{\mathbf{u}}_{OQ}$ (pull towards O), $+T\hat{\mathbf{u}}_{QP}$ (pull towards P) and the edge force $\mathbf{N}$. A smooth edge pushes only normally to itself, so the $\mathbf{j}_0$-component gives $-T\,\hat{\mathbf{u}}_{OQ}\cdot\mathbf{j}_0 + T\,\hat{\mathbf{u}}_{QP}\cdot\mathbf{j}_0 = 0$, that is $\hat{\mathbf{u}}_{QP}\cdot\mathbf{j}_0 = \sin\theta$, the statement of the hint.

Part (b). The $\mathbf{j}_0$-component of $\hat{\mathbf{u}}_{QP}$ is $\sin\theta$; the remainder has modulus $\sqrt{1-\sin^2\theta} = \cos\theta$ and lies in the plane normal to the edge, where a downward line making the angle ϕ with the vertical plane $x_0 = d$ has direction $\sin\phi\,\mathbf{i}_0 - \cos\phi\,\mathbf{k}_0$. Hence

$$\hat{\mathbf{u}} \equiv \hat{\mathbf{u}}_{QP} = \cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0, \qquad \|\hat{\mathbf{u}}\|^2 = \cos^2\theta\left(\sin^2\phi + \cos^2\phi\right) + \sin^2\theta = 1$$

Moving basis $\mathcal{B}_1 = \{\hat{\mathbf{u}}, \mathbf{e}_\theta, \mathbf{e}_\phi\}$. Differentiating $\hat{\mathbf{u}}$ with respect to θ and ϕ gives $\partial\hat{\mathbf{u}}/\partial\theta = \mathbf{e}_\theta$ and $\partial\hat{\mathbf{u}}/\partial\phi = \cos\theta\,\mathbf{e}_\phi$ with the unit vectors

$$\mathbf{e}_\theta = -\sin\theta\sin\phi\,\mathbf{i}_0 + \cos\theta\,\mathbf{j}_0 + \sin\theta\cos\phi\,\mathbf{k}_0, \qquad \mathbf{e}_\phi = \cos\phi\,\mathbf{i}_0 + \sin\phi\,\mathbf{k}_0$$

Change-of-basis matrix (columns are the components of $\hat{\mathbf{u}}, \mathbf{e}_\theta, \mathbf{e}_\phi$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta\sin\phi & -\sin\theta\sin\phi & \cos\phi \\ \sin\theta & \cos\theta & 0 \\ -\cos\theta\cos\phi & \sin\theta\cos\phi & \sin\phi \end{pmatrix}$$

- **Orthonormality ($R^TR = I$):** $\hat{\mathbf{u}}\cdot\hat{\mathbf{u}} = 1$; $\mathbf{e}_\theta\cdot\mathbf{e}_\theta = \sin^2\theta\sin^2\phi + \cos^2\theta + \sin^2\theta\cos^2\phi = 1$; $\mathbf{e}_\phi\cdot\mathbf{e}_\phi = 1$; $\hat{\mathbf{u}}\cdot\mathbf{e}_\theta = \sin\theta\cos\theta\left(1 - \sin^2\phi - \cos^2\phi\right) = 0$; $\hat{\mathbf{u}}\cdot\mathbf{e}_\phi = \cos\theta\sin\phi\cos\phi - \cos\theta\cos\phi\sin\phi = 0$; $\mathbf{e}_\theta\cdot\mathbf{e}_\phi = -\sin\theta\sin\phi\cos\phi + \sin\theta\cos\phi\sin\phi = 0$.
- **Determinant:** component by component, $\hat{\mathbf{u}}\times\mathbf{e}_\theta = \cos\phi\,\mathbf{i}_0 + \sin\phi\,\mathbf{k}_0 = \mathbf{e}_\phi$ (using $\sin^2\theta + \cos^2\theta = 1$), so the triad is right-handed and $\det[{}_0R_1] = +1$.

Derivatives of the moving basis (from the partial derivatives, $\dot{\mathbf{e}}_i = \dot\theta\,\partial\mathbf{e}_i/\partial\theta + \dot\phi\,\partial\mathbf{e}_i/\partial\phi$):

$$\dot{\hat{\mathbf{u}}} = \dot\theta\,\mathbf{e}_\theta + \dot\phi\cos\theta\,\mathbf{e}_\phi, \qquad \dot{\mathbf{e}}_\theta = -\dot\theta\,\hat{\mathbf{u}} - \dot\phi\sin\theta\,\mathbf{e}_\phi, \qquad \dot{\mathbf{e}}_\phi = \dot\phi\,(-\cos\theta\,\hat{\mathbf{u}} + \sin\theta\,\mathbf{e}_\theta)$$

For instance $\partial\mathbf{e}_\theta/\partial\theta = (-\cos\theta\sin\phi, -\sin\theta, \cos\theta\cos\phi) = -\hat{\mathbf{u}}$ and $\partial\mathbf{e}_\phi/\partial\phi = (-\sin\phi, 0, \cos\phi) = -\cos\theta\,\hat{\mathbf{u}} + \sin\theta\,\mathbf{e}_\theta$. The matrix $[{}_0R_1]^T[{}_0\dot R_1]$ built from these relations is skew-symmetric, as required for a rotating orthonormal basis.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom

Two: (θ, ϕ), as counted in Phase 1.

(b) Unit vector from Q to P

Obtained in Phase 2: $\hat{\mathbf{u}} = \cos\theta\sin\phi\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0 - \cos\theta\cos\phi\,\mathbf{k}_0$.

(c) Position, velocity and acceleration of Q and P

> **Pedagogical Justification:** Q slides along a straight edge, so only the chain rule on $\tan\theta$ is needed. P is Q plus a vector of variable length along a variable direction, so the product rule is needed.

Point Q (chain rule $\frac{d}{dt}\tan\theta = \dot\theta/\cos^2\theta$ and $\frac{d}{dt}\cos^{-2}\theta = 2\sin\theta\,\dot\theta/\cos^3\theta$):

$$\mathbf{r}_0^Q = d\,\mathbf{i}_0 + d\tan\theta\,\mathbf{j}_0, \qquad \mathbf{v}_0^Q = \frac{d\,\dot\theta}{\cos^2\theta}\,\mathbf{j}_0, \qquad \mathbf{a}_0^Q = \frac{d}{\cos^2\theta}\left(\ddot\theta + 2\tan\theta\,\dot\theta^2\right)\mathbf{j}_0$$

Position of P:

$$\mathbf{r}_0^P = \mathbf{r}_0^Q + \rho\,\hat{\mathbf{u}}, \qquad \rho = \ell - \frac{d}{\cos\theta}, \qquad \dot\rho = -\frac{d\sin\theta}{\cos^2\theta}\,\dot\theta$$

Velocity of P (product rule): $\mathbf{v}_0^P = \mathbf{v}_0^Q + \dot\rho\,\hat{\mathbf{u}} + \rho\,\dot{\hat{\mathbf{u}}}$. Project on $\mathcal{B}_1$ with $\mathbf{j}_0\cdot\hat{\mathbf{u}} = \sin\theta, \mathbf{j}_0\cdot\mathbf{e}_\theta = \cos\theta, \mathbf{j}_0\cdot\mathbf{e}_\phi = 0$:

- $\hat{\mathbf{u}}$-component: $\dfrac{d\dot\theta\sin\theta}{\cos^2\theta} + \dot\rho = \dfrac{d\dot\theta\sin\theta}{\cos^2\theta} - \dfrac{d\dot\theta\sin\theta}{\cos^2\theta} = 0.$
- $\mathbf{e}_\theta$-component: $\dfrac{d\dot\theta\cos\theta}{\cos^2\theta} + \rho\dot\theta = \dot\theta\left(\dfrac{d}{\cos\theta} + \ell - \dfrac{d}{\cos\theta}\right) = \ell\dot\theta.$
- $\mathbf{e}_\phi$-component: $0 + \rho\cos\theta\,\dot\phi = w\dot\phi.$

$$\boxed{\mathbf{v}_0^P = \ell\dot\theta\,\mathbf{e}_\theta + w\dot\phi\,\mathbf{e}_\phi}, \qquad \|\mathbf{v}_0^P\|^2 = \ell^2\dot\theta^2 + w^2\dot\phi^2, \qquad w = \ell\cos\theta - d$$

Acceleration of P. Differentiate with the product rule, using $\dot w = -\ell\sin\theta\,\dot\theta$ and the basis derivatives of Phase 2:

$$\frac{d}{dt}\left(\ell\dot\theta\,\mathbf{e}_\theta\right) = \ell\ddot\theta\,\mathbf{e}_\theta + \ell\dot\theta\left(-\dot\theta\,\hat{\mathbf{u}} - \dot\phi\sin\theta\,\mathbf{e}_\phi\right)$$

$$\frac{d}{dt}\left(w\dot\phi\,\mathbf{e}_\phi\right) = \left(\dot w\dot\phi + w\ddot\phi\right)\mathbf{e}_\phi + w\dot\phi^2\left(-\cos\theta\,\hat{\mathbf{u}} + \sin\theta\,\mathbf{e}_\theta\right)$$

Adding both and collecting terms (the $\mathbf{e}_\phi$ coefficient gathers $-\ell\sin\theta\,\dot\theta\dot\phi$ from the first line and $\dot w\dot\phi = -\ell\sin\theta\,\dot\theta\dot\phi$ from the second):

$$\boxed{\mathbf{a}_0^P = \left(-\ell\dot\theta^2 - w\cos\theta\,\dot\phi^2\right)\hat{\mathbf{u}} + \left(\ell\ddot\theta + w\sin\theta\,\dot\phi^2\right)\mathbf{e}_\theta + \left(w\ddot\phi - 2\ell\sin\theta\,\dot\theta\dot\phi\right)\mathbf{e}_\phi}$$

Components in $\mathcal{B}_0$ follow by multiplying the column of $\mathcal{B}_1$-components by $[{}_0R_1]$.

(d) Forces on P

> **Pedagogical Justification:** P is a free particle in contact only with the string, so the forces are its weight and the string tension. The edge force acts on the (massless) string at Q, not on P.

- Weight: $\mathbf{W} = -mg\,\mathbf{k}_0$. With $\mathbf{k}_0\cdot\hat{\mathbf{u}} = -\cos\theta\cos\phi$, $\mathbf{k}_0\cdot\mathbf{e}_\theta = \sin\theta\cos\phi$, $\mathbf{k}_0\cdot\mathbf{e}_\phi = \sin\phi$: $\mathbf{W} = mg\cos\theta\cos\phi\,\hat{\mathbf{u}} - mg\sin\theta\cos\phi\,\mathbf{e}_\theta - mg\sin\phi\,\mathbf{e}_\phi$.
- Tension: $\mathbf{T} = -T\,\hat{\mathbf{u}}$, directed towards Q, with $T \geq 0$ unknown.

(e) Equations of motion free of T

> **Pedagogical Justification:** Newton's second law $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{T}$ has the three unknowns $\ddot\theta, \ddot\phi, T$. Projecting on $\mathcal{B}_1$ isolates T in the $\hat{\mathbf{u}}$ equation, because $\mathbf{T}$ has no component along $\mathbf{e}_\theta$ or $\mathbf{e}_\phi$.

$$\hat{\mathbf{u}}:\ m\left(-\ell\dot\theta^2 - w\cos\theta\,\dot\phi^2\right) = mg\cos\theta\cos\phi - T$$

$$\mathbf{e}_\theta:\ m\left(\ell\ddot\theta + w\sin\theta\,\dot\phi^2\right) = -mg\sin\theta\cos\phi$$

$$\mathbf{e}_\phi:\ m\left(w\ddot\phi - 2\ell\sin\theta\,\dot\theta\dot\phi\right) = -mg\sin\phi$$

The last two contain only θ, ϕ and their derivatives; they are the requested pair (with $w = \ell\cos\theta - d$):

$$\boxed{\ddot{\theta} = -\frac{\sin\theta}{\ell}\left[(\ell\cos\theta - d)\,\dot{\phi}^2 + g\cos\phi\right]}, \qquad \boxed{\ddot{\phi} = \frac{2\ell\sin\theta\,\dot{\theta}\dot{\phi} - g\sin\phi}{\ell\cos\theta - d}}$$

The $\hat{\mathbf{u}}$ equation then gives the tension once the motion is known:

$$T = m\left[g\cos\theta\cos\phi + \ell\dot{\theta}^2 + (\ell\cos\theta - d)\cos\theta\,\dot{\phi}^2\right]$$

which is positive whenever $\cos\phi \geq 0$ and $w > 0$, consistent with a taut string.

Cross-check by Lagrange's equations. With kinetic energy $\tfrac{1}{2}m(\ell^2\dot{\theta}^2 + w^2\dot{\phi}^2)$ and $V = mgz_P$, $z_P = \mathbf{r}_0^P \cdot \mathbf{k}_0 = -\rho\cos\theta\cos\phi = -w\cos\phi$, the Lagrangian is $L = \tfrac{1}{2}m(\ell^2\dot{\theta}^2 + w^2\dot{\phi}^2) + mgw\cos\phi$. With $\partial w/\partial\theta = -\ell\sin\theta$:

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\theta}} - \frac{\partial L}{\partial\theta} = m\ell^2\ddot{\theta} + m\ell\sin\theta\left(w\dot{\phi}^2 + g\cos\phi\right) = 0$$

$$\frac{d}{dt}\frac{\partial L}{\partial\dot{\phi}} - \frac{\partial L}{\partial\phi} = m\left(w^2\ddot{\phi} + 2w\dot{w}\dot{\phi}\right) + mgw\sin\phi = 0$$

Dividing the first by $m\ell$ and the second by mw reproduces the two boxed equations. A symbolic comparison at three random states confirmed that the Lagrange expressions equal the $\mathbf{e}_\theta$ and $\mathbf{e}_\phi$ Newton equations multiplied by ℓ and w respectively (agreement to machine precision).

(f) Work of the tension

> **Pedagogical Justification:** The power of the tension on P is $\mathbf{T}\cdot\mathbf{v}_0^P$; it vanishes if $\mathbf{v}_0^P \perp \hat{\mathbf{u}}$.

$$P_T = \mathbf{T}\cdot\mathbf{v}_0^P = -T\,\hat{\mathbf{u}}\cdot\left(\ell\dot{\theta}\,\mathbf{e}_\theta + w\dot{\phi}\,\mathbf{e}_\phi\right) = 0$$

because $\hat{\mathbf{u}}$ is orthogonal to $\mathbf{e}_\theta$ and $\mathbf{e}_\phi$. Equivalently, the $\hat{\mathbf{u}}$-component of $\mathbf{v}_0^P$ in (c) cancelled: the string slides over the edge so that the hanging length grows at the rate at which Q moves along $\hat{\mathbf{u}}$. **No: the tension does no work on** P (official key, PDF page 76: $\hat{\mathbf{u}} \cdot \mathbf{v}_0^P = 0$).

(g) Mechanical energy

General expression (kinetic energy from (c); gravitational potential energy with Oz_0 up and $V = 0$ on the plate plane):

$$E = \tfrac{1}{2} m \|\mathbf{v}_0^P\|^2 + m g z_P = \tfrac{1}{2} m \left[\ell^2 \dot{\theta}^2 + (\ell \cos\theta - d)^2 \, \dot{\phi}^2 \right] - m g \, (\ell \cos\theta - d) \cos\phi$$

Initial value. At $t = 0$: $\tfrac{1}{2} m \|\mathbf{v}_0^P\|^2 = \tfrac{1}{2} m \ell g$, $w(0) = \ell \cos(\pi/3) - d = \ell/2 - d$ and $\cos\phi(0) = 1/2$:

$$E_0 = \tfrac{1}{2} m \ell g - m g \left(\frac{\ell}{2} - d \right) \frac{1}{2} = m g \left(\frac{\ell}{2} - \frac{\ell}{4} + \frac{d}{2} \right) = \frac{m g \, (\ell + 2d)}{4}$$

Conservation. By the work-energy theorem, $\frac{d}{dt} \tfrac{1}{2} m \|\mathbf{v}\|^2 = \mathbf{W} \cdot \mathbf{v} + \mathbf{T} \cdot \mathbf{v}$. The tension term is zero by (f), and $\mathbf{W} \cdot \mathbf{v} = -\frac{d}{dt}(m g z_P)$ because gravity is conservative. Hence E is **conserved**. Direct check, differentiating E and inserting the equations of (e):

$$\frac{\dot{E}}{m} = \ell^2 \dot{\theta} \ddot{\theta} + w^2 \dot{\phi} \ddot{\phi} + w \dot{w} \dot{\phi}^2 - g \dot{w} \cos\phi + g w \sin\phi \, \dot{\phi}$$

with $\ell^2 \dot{\theta} \ddot{\theta} = -\ell \sin\theta \, \dot{\theta} \left(w \dot{\phi}^2 + g \cos\phi \right)$, $w^2 \dot{\phi} \ddot{\phi} = w \dot{\phi} \left(2\ell \sin\theta \, \dot{\theta} \dot{\phi} - g \sin\phi \right)$ and $\dot{w} = -\ell \sin\theta \, \dot{\theta}$. The terms in $\ell w \sin\theta \, \dot{\theta} \dot{\phi}^2$ add to $-1 + 2 - 1 = 0$, the terms in $g \ell \sin\theta \, \dot{\theta} \cos\phi$ add to $-1 + 1 = 0$, and the terms in $g w \dot{\phi} \sin\phi$ add to $-1 + 1 = 0$, so $\dot{E} = 0$.

Angular momentum about O is not conserved. The torque of the tension about O has the $\mathbf{k}_0$-component $-T \, (\mathbf{r}_0^Q \times \hat{\mathbf{u}}) \cdot \mathbf{k}_0 = -T d \sin\theta \, (1 - \sin\phi)$, which is non-zero at $t = 0$, and the weight has a non-zero horizontal torque. No component of $\mathbf{H}_O$ is a first integral in general, so the energy is the conserved quantity asked about.

PHASE 4

Physical Interpretation & Verification

- **Numerical check** (Python, RK4 with step 10^{-4} s; $\ell = 2$ m, $d = 0.4$ m, $g = 9.81$ m/s^2, $\theta(0) = \phi(0) = \pi/3$, $\dot\phi(0) = 0$, hence $\dot\theta(0) = \quad g/\ell = 2.2147$ rad/s):

Quantity	Result
Initial energy per unit mass	$E_0/m = 6.8670$ J/kg, equal to $g(\ell + 2d)/4 = 6.8670$ J/kg
Energy drift over 4 s	$\max \lvert E - E_0\rvert/m = 6.5 \times 10^{-11}$ J/kg
Tension and hanging length	$\min T/m = 2.94$ m/s$^2 > 0$ and $\min \rho = 0.38$ m > 0
String-length constraint	$\lVert\mathbf{OQ}\rVert + \lVert\mathbf{QP}\rVert - \ell = 4 \times 10^{-16}$ m
Velocity formula of (c)	finite-difference velocity of $\mathbf{r}_0^P$ equals the closed form to four decimals
Angular momentum about O	components of $\mathbf{H}_O/m$ vary (the $\mathbf{i}_0$ component runs from -5.2 to 7.3 m^2/s)

- **Limit $d \to 0$:** $Q \to O$, $w = \ell\cos\theta$ and $E = \tfrac{1}{2}m\ell^2(\dot\theta^2 + \cos^2\theta\,\dot\phi^2) - mg\ell\cos\theta\cos\phi$, the energy of a spherical pendulum of length ℓ in these angular coordinates.
- **Geometry:** $\theta = 0$ puts Q at the foot of the perpendicular from O (distance d), where the hanging length $\ell - d$ is maximal. The equations are singular when $w = 0$ (zero hanging length), which the hypothesis of a taut string over the edge excludes.
- **Dimensions:** $[\ell\ddot\theta] = [g] = \mathrm{m/s^2}$, $[T] = \mathrm{kg\cdot m/s^2} = \mathrm{N}$, $[E_0] = \mathrm{kg\cdot m^2/s^2} = \mathrm{J}$.

`Worked Solution` *Source: Problems.pdf (PDF page 27); key (part f only) on PDF page 76*

PROBLEM 42 | Binet + Effective Potential

Particle Attracted to the Origin by $F = m\mu/r^{\alpha}$

Original statement · Problems.pdf, p. 29

(Advanced problem.) A point particle P of mass m is attracted by the origin of coordinates O of the inertial reference frame S_0 by a force with magnitude

$$F = \frac{m\mu}{r^{\alpha}}$$

where r is the distance between O and P, and μ, α are known positive constants.

Notation. The official text writes $F = m\alpha/r^{\beta}$ (strength α, exponent β) and asks for $\beta = 1, 2, 3, 4$. Here the strength is called μ and the exponent α, so the cases "$\alpha = 1, 2, 3, 4$" below are the official $\beta = 1, 2, 3, 4$, and μ/h^2 is the official α/h^2.

(a) Show that the angular momentum vector and the mechanical energy of P are conserved quantities of motion, and that the motion of P occurs in a plane.

(b) Using polar coordinates (r, θ) in the plane of motion, find a second-order differential equation to solve for $u \equiv 1/r$ as a function of θ (i.e., $u = u(\theta)$).

(c) Find a first integral of this equation and discuss the limits of motion of u as a function of μ/h^2, α, and the values of u and $u' \equiv du/d\theta$ at $t = 0$, where h is the magnitude of the angular momentum vector about the origin per unit mass. For $\alpha = 1, 2, 3,$ and 4, determine under which conditions: (i) the particle escapes to infinity ($u \to 0$); (ii) the particle reaches $r = 0$ ($u \to \infty$); (iii) the particle oscillates between two values of u.

Hint: try to solve first the problem with $\alpha = 2$, and then try the other values.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a particle P of mass m attracted to O by $\mathbf{F} = -\dfrac{m\mu}{r^{\alpha}}\mathbf{e}_r$.
- **Hypotheses:** S_0 inertial; $\mu > 0, \alpha > 0$ constants; no other forces; $h \neq 0$ unless stated.
- **Degrees of freedom:** free particle, $\mathrm{CDOF} = 3$; after part (a) the motion is planar, $\mathrm{CDOF} = 2$ with coordinates (r, θ).
- **Variables:** $u(\theta) = 1/r(\theta), u' = du/d\theta, h = \|\mathbf{r} \times \mathbf{v}\|$ [m^2/s].

Symbol	Meaning	SI unit
μ	force strength per unit mass, $F/m = \mu/r^{\alpha}$	$\mathrm{m}^{\alpha+1}/\mathrm{s}^2$
h	specific angular momentum	m^2/s
u	reciprocal radius	m^{-1}
ξ	specific mechanical energy	J/kg = $\mathrm{m}^2/\mathrm{s}^2$

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Change of basis (polar basis in the plane of motion).** Take $\mathbf{k}$ along $\mathbf{H}_O$. The polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

The angular velocity of $\mathcal{B}_1$ is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$; Poisson's formula $\dot{\mathbf{e}} = \boldsymbol{\omega} \times \mathbf{e}$ gives $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$. The product rule on $\mathbf{r} = r\mathbf{e}_r$ yields

$$\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta, \qquad \mathbf{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\mathbf{e}_\theta$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): conservation laws and planar motion

Pedagogical Justification: The force line of action passes through the fixed point O, so its torque about O vanishes, and the angular momentum theorem about a fixed point (Notes Eq. 4.53) gives a constant $\mathbf{H}_O$.

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = r\mathbf{e}_r \times \left(-\frac{m\mu}{r^\alpha}\mathbf{e}_r\right) = \mathbf{0} \implies \left.\frac{d\mathbf{H}_O}{dt}\right|_0 = \mathbf{0} \implies \mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v} = \text{const}$$

Since $\mathbf{r} \cdot \mathbf{H}_O \equiv 0$ and $\mathbf{v} \cdot \mathbf{H}_O \equiv 0$, P stays in the plane through O perpendicular to $\mathbf{H}_O$.

Pedagogical Justification: The force is purely radial and depends only on r, so its elementary work $\mathbf{F} \cdot d\mathbf{r} = F(r)\,dr$ is the exact differential of a function of r (equivalently $\nabla \times \mathbf{F} = \mathbf{0}$). The force is conservative, so the energy is conserved. The potential energy per unit mass follows from Barrow's rule with the reference at infinity (for $\alpha > 1$; for $\alpha \le 1$ the reference point is arbitrary):

$$V_0(r) = -\int_\infty^r \left(-\frac{\mu}{s^\alpha}\right) ds = \mu\left[\frac{s^{1-\alpha}}{1-\alpha}\right]_\infty^r = -\frac{\mu}{(\alpha-1)\,r^{\alpha-1}} \quad (\alpha > 1), \qquad V_0(r) = \mu \ln r \quad (\alpha = 1)$$

$$\xi = \tfrac{1}{2}v^2 + V_0(r) = \text{const}$$

Part (b): Binet equation for a power law

> **Pedagogical Justification:** Eliminate time with $h = r^2\dot\theta = \text{const}$, i.e. $\dot\theta = hu^2$. The chain rule for $r = u^{-1}$ gives $dr/d\theta = -u'/u^2$.

$$\dot r = \frac{dr}{d\theta}\dot\theta = \left(-\frac{u'}{u^2}\right)\left(hu^2\right) = -hu', \qquad \ddot r = \frac{d}{dt}\left(-hu'\right) = -hu''\,\dot\theta = -h^2u^2u''$$

The radial equation $\ddot r - r\dot\theta^2 = -\mu/r^\alpha = -\mu u^\alpha$ with $r\dot\theta^2 = \frac{1}{u}(hu^2)^2 = h^2u^3$ becomes

$$-h^2u^2u'' - h^2u^3 = -\mu u^\alpha \quad \overset{\div(-h^2u^2)}{\Longrightarrow} \quad \boxed{u'' + u = \frac{\mu}{h^2}\,u^{\alpha-2}}$$

Part (c): first integral

> **Pedagogical Justification:** Write the specific energy with $\dot r = -hu'$ and $r\dot\theta = hu$. Since ξ is constant, this is a first integral; it can be checked by multiplying the Binet equation by u' and integrating in θ.

$$\xi = \tfrac{1}{2}h^2\left[(u')^2 + u^2\right] + V_0\!\left(\frac{1}{u}\right) \implies \boxed{(u')^2 + W_{\text{eff}}(u) = E^*}, \quad E^* \equiv \frac{2\xi}{h^2}, \quad W_{\text{eff}}(u) = u^2 + \frac{2V_0(1/u)}{h^2}$$

Check: $\frac{d}{d\theta}\left[(u')^2 + W_{\text{eff}}\right] = 2u'u'' + W'_{\text{eff}}u' = 0$ gives $u'' = -\frac{1}{2}W'_{\text{eff}}$, which is the Binet equation. Motion exists only where $W_{\text{eff}}(u) \le E^*$; turning points ($u' = 0$) are the roots of $W_{\text{eff}}(u) = E^*$, and a minimum or a maximum of W_{eff} is a circular orbit.

Case $\alpha = 2$: inverse square (Kepler)

- Binet: $u'' + u = \mu/h^2$. Potential: $V_0 = -\mu u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2}u = \left(u - \dfrac{\mu}{h^2}\right)^2 - \left(\dfrac{\mu}{h^2}\right)^2$, a parabola with minimum $-(\mu/h^2)^2$ at $u = \mu/h^2$.
- **(i) Escape ($u \to 0$):** possible if $E^* \ge W_{\text{eff}}(0) = 0$, i.e. $\xi \ge 0$ (parabola or hyperbola).
- **(ii) Reaching $r = 0$:** impossible for $h \neq 0$, because $W_{\text{eff}} \to +\infty$ as $u \to \infty$.
- **(iii) Oscillation:** for $-(\mu/h^2)^2 < E^* < 0$, i.e. $-\mu^2/(2h^2) < \xi < 0$: u oscillates between the two roots (periapsis and apoapsis of an ellipse); $E^* = -(\mu/h^2)^2$ is the circular orbit.

Case $\alpha = 1$: $F \propto 1/r$

- Binet: $u'' + u = \dfrac{\mu}{h^2 u}$. Potential $V_0(1/u) = -\mu \ln u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2} \ln u$.
- $W_{\text{eff}} \to +\infty$ both as $u \to 0$ (because $-\ln u \to +\infty$) and as $u \to \infty$. Its minimum is at $2u - \dfrac{2\mu}{h^2 u} = 0$, i.e. $u_0 = \quad \mu/h$.
- **(i)** Escape is impossible for any finite energy. **(ii)** Reaching $r = 0$ is impossible. **(iii)** Every orbit with $E^* > W_{\text{eff}}(u_0)$ oscillates between two finite turning points; $E^* = W_{\text{eff}}(u_0)$ is the circular orbit $r = h/ \quad \mu$.

Case $\alpha = 3$: $F \propto 1/r^3$

- $V_0 = -\dfrac{\mu}{2r^2} = -\dfrac{\mu}{2} u^2$. Binet: $u'' + \lambda u = 0$ with $\lambda \equiv 1 - \mu/h^2$. Then $W_{\text{eff}} = \lambda u^2$ and $(u')^2 + \lambda u^2 = E^*$.
- $h^2 > \mu$ **($\lambda = \kappa^2 > 0$, $\kappa = \quad \sqrt{1 - \mu/h^2}$):** $u = A\cos(\kappa\theta + \delta)$ with $A = \quad \sqrt{u_0^2 + (u_0'/\kappa)^2}$. The cosine vanishes at a finite angle, so $u \to 0$ in every case: **(i) the particle always escapes**; a particle moving inward ($u_0' > 0$) is first reflected at the periapsis $u = A$. **(ii)** never, since $u \le A$. **(iii)** no oscillation between two positive values.
- $h^2 < \mu$ **($\lambda = -\kappa^2 < 0$, $\kappa = \quad \sqrt{\mu/h^2 - 1}$):** $u = C_1 e^{\kappa\theta} + C_2 e^{-\kappa\theta}$ with $C_1 = \frac{1}{2}(u_0 + u_0'/\kappa)$, $C_2 = \frac{1}{2}(u_0 - u_0'/\kappa)$. If $C_1 > 0$ (i.e. $u_0' > -\kappa u_0$), $u \to \infty$: **(ii) the particle reaches the origin**. If $C_1 < 0$ ($u_0' < -\kappa u_0$), u reaches 0 at a finite angle: **(i) escape**. If $C_1 = 0$, $u = u_0 e^{-\kappa\theta} \to 0$ only asymptotically (spiral $r = r_0 e^{\kappa\theta}$), a limiting case of escape. **(iii)** no oscillation.
- $h^2 = \mu$ **($\lambda = 0$):** $u'' = 0 \Rightarrow u = u_0 + u_0'\theta$ (reciprocal spiral). $u_0' > 0$: reaches the origin; $u_0' < 0$: escapes at $\theta = u_0/|u_0'|$; $u_0' = 0$: circle $r = 1/u_0$.

Case $\alpha = 4$: $F \propto 1/r^4$

- Binet: $u'' + u = \dfrac{\mu}{h^2} u^2$. $V_0 = -\dfrac{\mu}{3} u^3$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{3h^2} u^3 = u^2 \left(1 - \dfrac{2\mu}{3h^2} u\right)$.
- $W_{\text{eff}}'(u) = 2u - \dfrac{2\mu}{h^2} u^2 = 2u \left(1 - \dfrac{\mu}{h^2} u\right)$: stationary points $u = 0$ (minimum, $W = 0$) and $u_b = h^2/\mu$ (maximum, barrier $W_{\max} = u_b^2 - \dfrac{2\mu}{3h^2} u_b^3 = \dfrac{h^4}{3\mu^2}$). For $u > u_b$, W_{eff} decreases without bound and crosses zero at $u_0 = 3h^2/(2\mu) > u_b$.
- **There is no interior minimum**, so the allowed set $\{W_{\text{eff}} \le E^*\}$ either contains $u = 0$ or extends to $u \to \infty$. Because $\dot\theta = h u^2 > 0$, u' has the sign of $\dot u$.
- $0 \le E^* < W_{\max}$**:** two roots $u_1 \in [0, u_b)$ and $u_2 > u_b$; allowed $u \le u_1$ or $u \ge u_2$. If $u(0) \le u_1$ the particle is reflected at u_1 and escapes; if $u(0) \ge u_2$ it is reflected at u_2 and falls to the origin.
- $E^* < 0$**:** only $u \ge u_2 > u_0$ is allowed; the particle reaches $r = 0$. $E^* > W_{\max}$**:** no turning points; $u'(0) < 0$ (moving outward) escapes and $u'(0) > 0$ (moving inward) falls to the origin. $E^* = W_{\max}$**:** the unstable circular orbit $u = u_b$ and asymptotic approaches to it.

- **(i)** escape: $0 \le E^* < W_{\max}$ with $u(0) < u_b$, or $E^* > W_{\max}$ with $u'(0) < 0$. **(ii)** reaching $r = 0$: $E^* < 0$, or $0 \le E^* < W_{\max}$ with $u(0) > u_b$, or $E^* > W_{\max}$ with $u'(0) > 0$. **(iii)** bounded oscillation between two values of u: **does not occur**; it would need an interior minimum of W_{eff}, and the only bounded motion is the unstable circle $u = u_b$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional check:** $[F/m] = \mathrm{m/s^2} = [\mu]/\mathrm{m}^\alpha$ gives $[\mu] = \mathrm{m}^{\alpha+1}/\mathrm{s}^2$. With $[h^2] = \mathrm{m}^4/\mathrm{s}^2$ and $[u] = \mathrm{m}^{-1}$, $\dfrac{\mu}{h^2}u^{\alpha-2}$ has units $\mathrm{m}^{\alpha-3}\,\mathrm{m}^{-(\alpha-2)} = \mathrm{m}^{-1}$, the same as u and u'' (θ is dimensionless).
- **Stability of circular orbits (consequence of the first integral):** for general α, $W'_{\text{eff}}(u_0) = 0$ gives $u_0^{\alpha-3} = h^2/\mu$ and $W''_{\text{eff}}(u_0) = 2(3-\alpha)$, so the circular orbit is a minimum (stable) for $\alpha < 3$ and a maximum (unstable) for $\alpha > 3$, in agreement with the cases above ($\alpha = 4$: barrier at $u_b = h^2/\mu$).
- **Numerical check** (Python, RK4 with step 10^{-4} rad on $u'' = -u + (\mu/h^2)u^{\alpha-2}$, units with $\mu = h = 1$ unless stated):

Case	Initial data (u_0, u_0')	Predicted	Integration result
$\alpha = 2$	$(1.5, 0)$	bounded, 0.5 to 1.5	$u \in [0.5000, 1.5000]$ over 60 rad
$\alpha = 2$	$(0.5, -0.9)$, $E^* > 0$	escape	$u = 0$ at $\theta = 0.823$
$\alpha = 4$	$(0.5, 0)$, $E^* = 1/6 < 1/3$	reflected, then escape	$u = 0$ at $\theta = 2.078$
$\alpha = 4$	$(1.2, 0)$, $E^* = 0.288 < 1/3$	collapse	$u \to \infty$ at $\theta = 3.713$
$\alpha = 4$	$(0.5, 0.9)$, $E^* > 1/3$	collapse	$u \to \infty$ at $\theta = 3.258$
$\alpha = 4$	$(0.5, -0.9)$, $E^* > 1/3$	escape	$u = 0$ at $\theta = 0.524$
$\alpha = 4$	$(1, 0)$, $E^* = W_{\max}$	unstable circular orbit	$u = 1$ for 20 rad
$\alpha = 3$, $\mu/h^2 = 2$	$(1, 0.2)$, $C_1 > 0$	collapse	$u = \cosh\theta + 0.2\sinh\theta$ is finite at every finite θ and $\to \infty$ only as $\theta \to \infty$ (the particle reaches $r = 0$ in finite time after infinitely many turns); $\theta = 4.423$ is the integrator cutoff
$\alpha = 3$, $\mu/h^2 = 2$	$(1, -1.5)$, $C_1 < 0$	escape	$u = 0$ at $\theta = 0.805$
$\alpha = 3$, $\mu/h^2 = 1/2$	$(1, 0)$	escape	$u = 0$ at $\theta = 2.222$
$\alpha = 3$, $\mu = h^2$	$(1, -0.1)$	escape at $\theta = 10$	$u = 0$ at $\theta = 10.000$
$\alpha = 1$	$(0.5, 0.3)$	bounded	$u \in [0.471, 1.653]$ over 100 rad

The official key (Problems.pdf) gives no answer for Problem 42, so these checks are against the derived classification only.

Worked Solution

Source: Problems.pdf (PDF page 29); no key available

PROBLEM 43 | Angular Momentum + Energy

Two Connected Particles through a Hole

Original statement · Problems.pdf, p. 30

We consider a smooth horizontal plane with a small hole in it. One of the ends of a massless cord of length $2a$ is passed through the hole and then, two point particles, P_1 and P_2, each with the same mass m, are attached to the ends of the cord. Therefore, P_1 is constrained to move on the plane and P_2 moves along a vertical line below the hole.

Initially, half of the cord lies under the plane and half of the cord is on the plane (so $r(0) = a$). At that instant, P_1 is pushed with a speed v_0 perpendicular to the cord, counterclockwise as seen from above. Assume that the cord is always in tension.

(a) What is the value of v_0 if P_1 describes a circumference?

(b) In that case, what is the value of the tension in the cord?

(c) If $v_0 = \sqrt{8ag/3}$, reduce the problem to integrals (i.e., quadratures).

(d) Show that for this value of v_0, P_1 oscillates between $r = a$ and $r = 2a$.

(e) Apply Newton's law to P_2 and show that the tension of the cord is $T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$.

(f) Write down the new equations of motion if the plane on which P_1 moves is rough, with a friction coefficient μ.

Hint: the roots of the polynomial $-3x^3 + 7x^2 - 4$ *are* $x_1 = 1$, $x_2 = 2$ *and* $x_3 = -2/3$.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** P_1 on a smooth horizontal plane at polar position (r, θ) and P_2 hanging on the vertical through the hole O, joined by an inextensible massless cord of length $2a$ passing through O.
- **Constraints and degrees of freedom:** the cord length gives $r + \ell_2 = 2a$, where ℓ_2 is the hanging length, so $z_2 = -\ell_2 = r - 2a$ (Oz up). The generalized coordinates are (r, θ), $\mathrm{CDOF} = 2$.
- **Hypotheses:** inertial frame S_0 with O at the hole; smooth hole and plane for parts (a)-(e); the cord stays taut; no drag.
- **Initial state:** $r(0) = a, \dot{r}(0) = 0, r(0)\dot{\theta}(0) = v_0$, hence $\dot{\theta}(0) = v_0/a$ and $\dot{z}_2(0) = 0$.

Symbol	Meaning	SI unit
m	mass of each particle	kg
a	half of the cord length	m
v_0	initial speed of P_1	m/s
g	gravitational acceleration	m/s^2
μ	friction coefficient (part f)	1

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, horizontal plane $z = 0$, $\mathbf{k}$ up.
- **Change of basis (polar basis of P_1).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

with $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$, $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$ (Poisson).

- **Kinematics:** $\mathbf{r}_1 = r\mathbf{e}_r$, $\mathbf{v}_1 = \dot r\mathbf{e}_r + r\dot\theta\mathbf{e}_\theta$; $\mathbf{r}_2 = (r-2a)\mathbf{k}$, $\mathbf{v}_2 = \dot r\mathbf{k}$.

$$T_{\text{kin}} = \tfrac{1}{2}m\left(\dot r^2 + r^2\dot\theta^2\right) + \tfrac{1}{2}m\dot r^2 = m\dot r^2 + \tfrac{1}{2}mr^2\dot\theta^2, \qquad V = mgz_2 = mg(r-2a)$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Circular motion, $r \equiv a$

Pedagogical Justification: On the plane the only horizontal force on P_1 is the tension $-T\mathbf{e}_r$, which is central. For $r = a$ the radial acceleration is purely centripetal; P_2 must stay at rest.

$$P_1:\ m\left(\ddot r - r\dot\theta^2\right) = -T \quad \overset{r=a,\,\ddot r=0}{} \quad T = ma\dot\theta^2 = \frac{mv_0^2}{a}, \qquad P_2:\ m\ddot z_2 = T - mg = 0 \implies T = mg$$

$$\frac{mv_0^2}{a} = mg \implies v_0 = \sqrt{ag}$$

(b) Tension in the circular case

$$T = mg$$

(c) Quadratures for $v_0 = \sqrt{8ag/3}$

Pedagogical Justification: The tension on P_1 is central, so $h = r^2\dot\theta$ is constant; the plane and hole are smooth and the cord inextensible, so $E = T_{\text{kin}} + V$ is constant. The two first integrals leave one equation for $r(t)$.

$$h = r^2\dot\theta = av_0 = a\quad\frac{8ag}{3} \implies h^2 = \frac{8}{3}a^3 g$$

$$E = m\dot r^2 + \frac{mh^2}{2r^2} + mg(r-2a), \qquad E(0) = \tfrac{1}{2}m\cdot\frac{8ag}{3} - mga = \frac{1}{3}mga$$

Equating, dividing by m and isolating $\dot r^2$:

$$\dot r^2 + \frac{4a^3 g}{3r^2} + gr - 2ga = \frac{1}{3}ga \implies \dot r^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2} = \frac{g}{3r^2}\left(7ar^2 - 3r^3 - 4a^3\right)$$

Separating variables and applying Barrow's rule from $r(0) = a$:

$$\frac{dr}{dt} = \pm\quad\frac{g}{3}\,\frac{-3r^3 + 7ar^2 - 4a^3}{r} \implies \int_a^r \frac{r'\,dr'}{-3r'^3 + 7ar'^2 - 4a^3} = \pm\quad\frac{g}{3}\,t$$

(d) Oscillation between $r = a$ and $r = 2a$

> **Pedagogical Justification:** Motion requires $\dot r^2 \geq 0$, i.e. $P(r) = -3r^3 + 7ar^2 - 4a^3 \geq 0$. With $x = r/a$ the polynomial is $a^3(-3x^3 + 7x^2 - 4)$, and the hint gives its roots.

$$-3x^3 + 7x^2 - 4 = -(x-1)(x-2)(3x+2)$$

Check of the roots: $x = 1$: $-3 + 7 - 4 = 0$; $x = 2$: $-24 + 28 - 4 = 0$; $x = -2/3$: $\frac{8}{9} + \frac{28}{9} - 4 = 0$. Since $r > 0$, $3x + 2 > 0$ and $P \geq 0$ requires $(x-1)(x-2) \leq 0$:

$$1 \leq x \leq 2 \iff a \leq r(t) \leq 2a$$

The turning points ($\dot r = 0$) are $r_{\min} = a$ and $r_{\max} = 2a$, and the motion oscillates between them.

(e) Tension of the cord

> **Pedagogical Justification:** Apply Newton's second law to P_2 (weight and tension) with $\ddot z_2 = \ddot r$, because $z_2 = r - 2a$.

$$m\ddot r = T - mg \implies T = m\left(g + \ddot r\right)$$

Differentiate $\dot r^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2}$ with respect to time (chain rule, $\frac{d}{dt} r^{-2} = -2r^{-3}\dot r$):

$$2\dot r \ddot r = \left(-g + \frac{8a^3 g}{3r^3}\right)\dot r \implies \ddot r = -\frac{g}{2} + \frac{4a^3 g}{3r^3}$$

$$\boxed{T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)}$$

$T > 0$ for all $r \in [a, 2a]$, so the cord stays taut.

(f) Rough plane

> **Pedagogical Justification:** Coulomb friction acts on P_1 opposite to its velocity, with magnitude μN_1. The normal force is $N_1 = mg$, because the cord is horizontal on the plane and only the weight acts vertically.

$$\mathbf{F}_f = -\mu mg\,\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = -\mu mg\,\frac{\dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta}{\dot r^2 + r^2\dot\theta^2}$$

$$\mathbf{e}_r:\ m\left(\ddot r - r\dot\theta^2\right) = -T - \mu m g\frac{\dot r}{\dot r^2 + r^2\dot\theta^2}$$

$$\mathbf{e}_\theta:\ m\left(r\ddot\theta + 2\dot r\dot\theta\right) = -\mu m g\frac{r\dot\theta}{\dot r^2 + r^2\dot\theta^2}, \qquad P_2:\ m\ddot r = T - mg$$

Adding the $\mathbf{e}_r$ equation and the P_2 equation eliminates T; multiplying the $\mathbf{e}_\theta$ equation by r gives the evolution of h:

$$2\ddot r - r\dot\theta^2 = -g - \mu g\frac{\dot r}{\dot r^2 + r^2\dot\theta^2}, \qquad \frac{d}{dt}\left(r^2\dot\theta\right) = -\mu g\,\frac{r^2\dot\theta}{\dot r^2 + r^2\dot\theta^2}$$

Angular momentum and energy are no longer conserved (friction dissipates energy), so the problem cannot be reduced to the quadratures of (c).

PHASE 4

Physical Interpretation & Dimensional Verification

- **Tension at the turning points:** $T(a) = mg\left(\tfrac12 + \tfrac43\right) = \tfrac{11}{6}mg$ and $T(2a) = mg\left(\tfrac12 + \tfrac{4}{24}\right) = \tfrac23 mg$, both positive.
- **Consistency with (a):** for $v_0^2 = ag$, $h^2 = a^3 g$ and $\ddot r(a) = -\tfrac{g}{2} + \tfrac{h^2}{2a^3} = 0$, so $r = a$ is an equilibrium of the radial equation, as expected for the circular case.
- **Radial period** (consequence of (c) and (d)): with $x = r/a$ and $x = \tfrac32 - \tfrac12\cos s$, $(x-1)(2-x) = \tfrac14\sin^2 s$ and $dx = \tfrac12\sin s\,ds$, so $\tau_r = 2\sqrt{\dfrac{3a}{g}}\displaystyle\int_0^{\pi}\frac{x(s)\,ds}{3x(s)+2} \approx 6.342\sqrt{\dfrac{a}{g}}$.
- **Numerical check** (Python, RK4 on $\ddot r = -g/2 + 4a^3 g/(3r^3)$ with $a = g = 1$, $r(0) = 1$, $\dot r(0) = 0$): maximum radius 2.0000, period 6.3425 s (the quadrature gives 6.3424), and the energy-integral residual stays below 4×10^{-14}.
- **Dimensions:** $[T] = [m][g] = \mathrm{N}$ and $[v_0] = \sqrt{[a][g]} = \mathrm{m/s}$.
- **Key:** the official key (Problems.pdf) has no entry for Problem 43.

Worked Solution

Source: Problems.pdf (PDF page 30); no key available

PROBLEM 44 | Angular Momentum + Vis-Viva

The Little Prince Throwing Seeds

Original statement · Problems.pdf, p. 31

The little Prince is on his spherical asteroid, which has radius R and mass M, and does not rotate. While he sits at the equator, he throws a baobab seed with an initial velocity of magnitude v_0 and angle $\beta = 60^\circ$ with respect to the local horizontal. Knowing that it reaches an apocenter of $r_a = 3R$ before falling back, determine v_0 as a function of the parameters of the problem and the universal gravitational constant G.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a seed of mass $m \ll M$ launched from the surface ($r_0 = R$) with speed v_0 at 60° above the local horizontal; it reaches the apocenter $r_a = 3R$.
- **Hypotheses:** spherically symmetric, non-rotating asteroid, so the gravitational force $\mathbf{F} = -\dfrac{GMm}{r^2}\mathbf{e}_r$ is central and conservative with $\mu = GM$; vacuum (no drag); inertial frame S_0 centred on the asteroid.
- **Degrees of freedom:** free particle, planar motion by central-force conservation of $\mathbf{H}_O$; coordinates (r, θ).
- **Initial state:** $r = R$, radial component $v_{r0} = v_0 \sin 60^\circ = \frac{3}{2} v_0$ and transverse component $v_{\theta 0} = v_0 \cos 60^\circ = \frac{1}{2} v_0$.

Symbol	Meaning	SI unit
M, R	mass and radius of the asteroid	kg, m
G	universal gravitational constant, 6.6743×10^{-11}	$\mathrm{m^3/(kg\ s^2)}$
$\mu = GM$	gravitational parameter	$\mathrm{m^3/s^2}$
β	launch angle above the horizontal	60°
r_a	apocenter radius	$3R$

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Change of basis (polar basis in the orbital plane).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by the true anomaly θ about $\mathbf{k}$ (along $\mathbf{H}_O$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

- **Velocity in $\mathcal{B}_1$:** $\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta$; at launch $\dot{r} = v_0 \sin 60^\circ$ and $r\dot{\theta} = v_0 \cos 60^\circ$. At the apocenter $\dot{r} = 0$, so the velocity is purely transverse.

PHASE 3

Step-by-Step Mathematical Deduction

1. Conservation of specific angular momentum

Pedagogical Justification: The gravitational force passes through O, so $\mathbf{M}_O = \mathbf{0}$ and $h = \|\mathbf{r} \times \mathbf{v}\| = r^2\dot{\theta}$ is constant. Evaluate it at launch and at the apocenter, where $v_a = r_a\dot{\theta}_a$.

$$h = R\,(v_0 \cos 60^\circ) = \frac{1}{2}Rv_0, \qquad h = r_a v_a = 3Rv_a \implies v_a = \frac{v_0}{6}$$

2. Conservation of specific energy

Pedagogical Justification: Gravity is conservative with $V_0 = -\mu/r$ (derived with Barrow's rule, reference at infinity), so $\xi = \frac{1}{2}v^2 - GM/r$ takes the same value at launch and at the apocenter.

$$\frac{1}{2}v_0^2 - \frac{GM}{R} = \frac{1}{2}\left(\frac{v_0}{6}\right)^2 - \frac{GM}{3R} = \frac{v_0^2}{72} - \frac{GM}{3R}$$

3. Solving for v_0

$$\left(\frac{1}{2}-\frac{1}{72}\right)v_0^2=\left(1-\frac{1}{3}\right)\frac{GM}{R}\implies\frac{35}{72}v_0^2=\frac{2}{3}\frac{GM}{R}\implies v_0^2=\frac{2}{3}\cdot\frac{72}{35}\frac{GM}{R}=\frac{48}{35}\frac{GM}{R}$$

$$\boxed{v_0=\sqrt{\frac{48}{35}\frac{GM}{R}}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Bound orbit:** the escape speed at the surface is $v_e=\sqrt{2GM/R}$, i.e. $v_e^2=\frac{70}{35}GM/R$. Since $\frac{48}{35}<2$, $v_0<v_e$ and the orbit is an ellipse: the seed falls back.
- **Orbital parameters:** $h^2=\frac{1}{4}R^2v_0^2=\frac{12}{35}GMR$, $p=h^2/\mu=\frac{12}{35}R$. From $r_a=p/(1-e)=3R$: $1-e=\frac{4}{35}$ and $e=\frac{31}{35}\approx0.8857$. The periapsis of the full conic is $r_p=p/(1+e)=\frac{2}{11}R<R$, inside the asteroid; only the arc from the launch point to the apocenter is physical, and the seed lands before reaching the periapsis.
- **Independent check with the Binet solution.** With $u=\frac{\mu}{h^2}(1+e\cos\theta)$ and the launch values $u(\theta_0)=1/R$, $u'(\theta_0)=-\dot r/h$ (here $\dot r=\frac{3}{2}v_0$), the eccentricity formula of the theory page gives

$$e=\frac{h^2}{\mu}\sqrt{\left(\frac{1}{R}-\frac{\mu}{h^2}\right)^2+\left(\frac{\dot r}{h}\right)^2}=\frac{12R}{35}\sqrt{\frac{529}{144R^2}+\frac{3}{R^2}}=\frac{12R}{35}\cdot\frac{31}{12R}=\frac{31}{35}$$

and the launch angle follows from $\tan\beta=\dot r/(r\dot\theta)=\frac{R}{p}\,e\sin\theta_0$ with $\cos\theta_0=\frac{p/R-1}{e}=-\frac{23}{31}$, $\sin\theta_0=\frac{\sqrt{432}}{31}$:

$$\tan\beta=\frac{35}{12}\cdot\frac{31}{35}\cdot\frac{\sqrt{432}}{31}=\frac{\sqrt{432}}{12}=\sqrt{3}\implies\beta=60^\circ$$

- **Numerical check** (Python, velocity-Verlet integration of $\ddot{\mathbf{r}} = -\mu\mathbf{r}/r^3$ with $\mu = R = 1$ and $v_0 = \sqrt{48/35}$): the maximum radius reached is $2.99999999981R$, in agreement with $r_a = 3R$.
- **Dimensions:** $[v_0] = \sqrt{[G][M]/[R]} = \sqrt{\dfrac{\mathrm{m}^3}{\mathrm{kg\,s}^2}\cdot\dfrac{\mathrm{kg}}{\mathrm{m}}} = \mathrm{m/s}$.
- **Key:** the official key (Problems.pdf) has no entry for Problem 44.

Worked Solution

Source: Problems.pdf (PDF page 31); no key available

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Study notes written from the official course material (Problems.pdf · p. 9). Online version: subjects/engineering-mechanics/problemas/topic-4.html